A0 dense causal Transformer

Task and source

The Phase 1 task is 128-word legal continuation given a 512-word prompt. The four official sources are Acts, High Court, Supreme Court, and Tribunal material. This report is a later reconstruction based on recovered source and archived run evidence, not an executed training notebook.

Architecture

A 12-block decoder-only Transformer predicts the next subword token from a causal context. Width 768, 12 attention heads of 64, 2048-wide SwiGLU feed-forward layers, RMSNorm, rotary positions, and tied input/output embeddings make the 110.1M-parameter graph.

Stage	512-word prompt -> Library-trained BPE -> 12 causal blocks
Continuation	RMSNorm / RoPE / SwiGLU -> Vocabulary logits -> 128-word continuation

Token and optimization science

The model learns by minimizing next-token cross-entropy under causal masking. An attention head compares query/key vectors, obtains a weighted value mixture, and updates the residual stream; rotary phases give position-sensitive comparisons. Decoding then turns logits into a 128-word continuation. Tokenizer choice affects sequence length, vocabulary utilization, and the optimization budget. The original source config and scripts identify the actual optimizer, schedule, context and generation policy.

Reported outcome

0.8304 (historical Codabench screenshot). Historical screenshot numbers are not freshly retrieved leaderboard results. Archived local metric files, when present, provide separately measured BERTScore, ROUGE-L and BLEU-4; no metric is invented for runs without such evidence.

Reproduction boundary

The code package contains the recovered model, tokenizer, training, evaluation and export scripts, plus original small logs/metrics. The complete ZIP holds selected checkpoint weights and official data, which are too large or inappropriate to duplicate in GitHub. Re-execution requires their original hashes and path remapping.

Configured dimensions

The following values are read directly from the original configuration. A continued-training config may inherit the parent model graph; that relationship is identified in the README and source code.

Setting	Archived value
vocab_size	32768
max_seq_len	2048
n_layers	12
d_model	768
n_heads	12
d_head	64
d_ff	2048
rope_theta	10000.0
dropout	0.0
gradient_checkpointing	True

Archived evaluations

This table is a small sample of per-policy and per-checkpoint records, not a single comparable leaderboard ranking. Full machine-readable values and names are in evidence/archived_metric_index.json.

Metric file	BERTScore F1	ROUGE-L F1	BLEU-4
No preserved metric JSON	—	—	—

Evidence count and caveat

Copied 6 compact log/metric/manifest artifacts from CS781_A0_Complete_20261006.zip. They support the run lineage and measured local evaluations. A lower generation loss need not imply better BERTScore or legal accuracy; checkpoint and decoding policies must be evaluated separately. Original October 6 source archive contains no .ipynb file, so a later illustrated notebook is explicitly marked as reconstructed.